

Bernard Lee

DATA TECHNOLOGY & AI LEADER

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Technology leader spanning enterprise data architecture, AI, analytics, and digital delivery. Combines hands-on engineering with architecture governance, business-case development, and cross-functional leadership—translating complex business needs into secure, scalable platforms with measurable operational and commercial outcomes.

SELECTED IMPACT

10–15 project architectures reviewed for secure, scalable design	5h→<1h external-data retrieval accelerated through Python pipelines	+25 pts health-program adoption across enterprise clients	25M financial communications processed by an NLP framework
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EXPERIENCE HIGHLIGHTS

Hong Kong Monetary Authority Jul 2025–Present
Technology Innovation Lead

- Re-architected a four-service Azure research solution into one containerized AWS web application within a month, supporting four data sources and enabling planned Azure decommissioning.
- Built a natural-language analytics PoC in three months, orchestrating schema discovery, RAG, SQL generation, validation, execution, and AI visualization for permission-controlled data.
- Led the Architecture Review Board-approved design of a Kong enterprise API and AI gateway intended to support up to 100M transactions, coordinating vendor teams of 15–20 engineers.

Hong Kong Monetary Authority Nov 2023–Jun 2025
Data Architect

- Reviewed data flows and proposals for 10–15 projects and defined the approved enterprise medallion reference architecture.
- Established secure web-scraping patterns adopted by at least five teams and built ingestion pipelines that cut retrieval from five hours to under one.
- Delivered architecture and AI knowledge-sharing sessions for audiences ranging from 25 to more than 300 people.

AIA Hong Kong & Macau Jan 2019–Jul 2022
Associate Director · Senior Manager, Health & Wellness

- Delivered five digital health programs for three enterprise clients covering 6,000+ employees, increasing adoption by 25 percentage points.
- Productionized 90%-accurate OCR for scoped claims, replacing 25% of manual quality-assurance workload.
- Built business cases covering US\$3M in analytics capability, US\$1M in application development, and multiple PoCs.

PROFILE

Sheffield Hallam University
BEng Electrical & Electronics Engineering
First City University College · 2010
Hong Kong Permanent Resident
English · Cantonese

Additional proof points
Reduced BI queries from minutes to seconds across global finance dashboards. Built a Python NLP framework for 25M Bloomberg, Reuters, and email messages; reuse cut a later investigation's delivery time by 67%.

Core stack

Python

SQL

TypeScript

Next.js

FastAPI

AWS

Azure

Databricks

Power BI

Tableau

D3.js

Kong

Docker

Kubernetes

LLMs

RAG

Best-fit focus
Enterprise data architecture · AI strategy & delivery · Financial services · Analytics platforms · Solution architecture · Digital transformation